

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

# Event Management Platform

Detailed project brief generated from the project record.

CANDIDATE

Varun Chopra

DURATION

14 days

TECHNOLOGY STACK

Java

Spring Boot

Angular

TypeScript

REST API

MySQL

## Project Overview

Summary of the problem, purpose and expected outcome.

**PROJECT SUMMARY**

A web app where organizers can create public events and users can browse and register. Includes admin tools for event moderation.

|                                              |                                      |                                        |
|----------------------------------------------|--------------------------------------|----------------------------------------|
| <div>CANDIDATE</div> <div>Varun Chopra</div> | <div>TYPE</div> <div>fullstack</div> | <div>DURATION</div> <div>14 days</div> |
|----------------------------------------------|--------------------------------------|----------------------------------------|

**Primary Objectives**

- Implement a secure REST API with Spring Boot.
- Build a responsive Angular UI for all user roles.
- Design a relational schema in MySQL.
- Deliver automated tests and a CI pipeline.

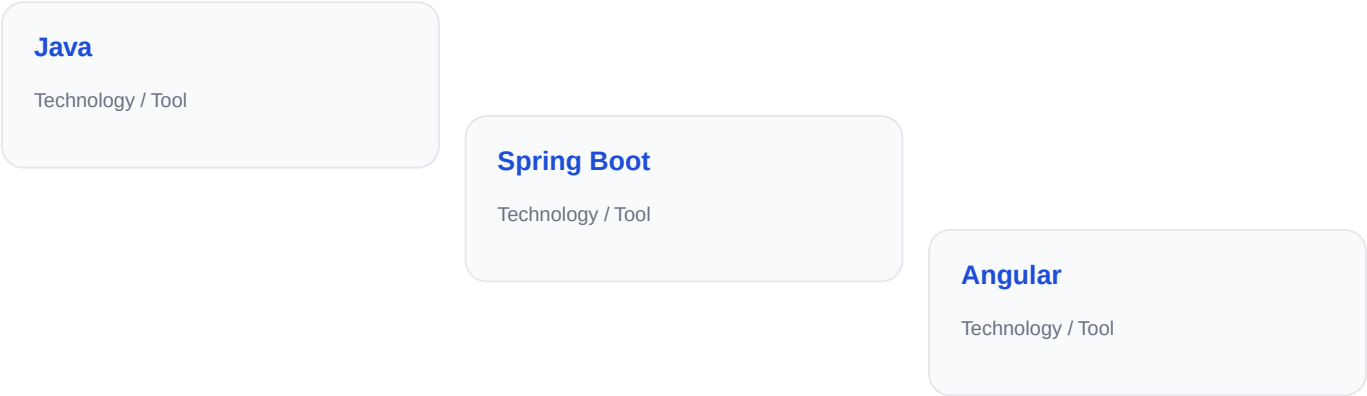
## Candidate Details

NAME

Varun Chopra

## Technology Stack

Technologies, frameworks and tools expected for implementation.



### TypeScript

Technology / Tool

### REST API

Technology / Tool

### MySQL

Technology / Tool

## Functional Requirements

- Register new users with email verification.
- Login with JWT token authentication.
- Create, read, update, delete (CRUD) events.
- List events with pagination and search.
- Register for an event and view personal registrations.
- Admin can view event attendance reports.

## Non-Functional Requirements

- Passwords stored with bcrypt hashing.
- All inputs validated server side.
- API responses under 200 ms for 95% of requests.
- Responsive UI for desktop and mobile.
- Data consistency enforced via foreign keys.

## Modules & Features

Major functional areas that make up the project.

### Authentication

Handles user sign up, login, JWT issuance.

- Email registration
- Login
- Password reset

### Event Management

CRUD operations for events.

- Create event
- Edit event
- Delete event
- Search & filter

### Registration

User enrollment in events.

- Register for event

- Cancel registration
- View my registrations

## Admin Dashboard

Admin overview and reporting.

- View all events
- View registration stats
- Moderate content

## Implementation Roadmap

Recommended phases and milestones for completing the project.

1

### Foundation

Initialize Git repo with Maven and Angular CLI, Configure CI (GitHub Actions) for build & test, Create Dockerfile for backend

- Initialize Git repo with Maven and Angular CLI
- Configure CI (GitHub Actions) for build & test
- Create Dockerfile for backend

2

### Backend Core

Design MySQL schema (users, events, registrations), Implement User, Event entities with JPA, Create AuthController (register, login), Create EventController (CRUD), Add JWT filter

- Design MySQL schema (users, events, registrations)
- Implement User, Event entities with JPA
- Create AuthController (register, login)
- Create EventController (CRUD)
- Add JWT filter

3

### Frontend Core

Set up Angular project with routing, Create AuthService (login, register), Build Login & Register components, Build EventList component with pagination, Integrate JWT interceptor

- Set up Angular project with routing
- Create AuthService (login, register)
- Build Login & Register components
- Build EventList component with pagination
- Integrate JWT interceptor

4

Intermediate Features

Implement RegistrationController (register, cancel), Add MyRegistrations component, Create AdminDashboard component, Role based route guards, Unit tests for services

- Implement RegistrationController (register, cancel)
- Add MyRegistrations component
- Create AdminDashboard component
- Role based route guards
- Unit tests for services

5

Testing & Deployment

Write integration tests for API endpoints, Add end to end Cypress tests for UI, Configure Docker Compose (backend, db, frontend), Run performance smoke test, Update README with deployment steps

- Write integration tests for API endpoints
- Add end to end Cypress tests for UI
- Configure Docker Compose (backend, db, frontend)
- Run performance smoke test
- Update README with deployment steps

API Design

Key API endpoints and expected responsibilities.

POST

/api/auth/register

Create new user

POST

/api/auth/login

Authenticate and return JWT

GET

/api/events

List events with filters

POST

/api/events

Create event (organizer)

POST

/api/registrations

Register current user for event

GET

/api/admin/reports

Admin attendance statistics

### Evaluation Criteria

Criteria that can be used to assess the completed project.

- Criterion: Correctness | Description: All functional requirements work as specified. | Weight: 30
- Criterion: Code Quality | Description: Clean, modular code with proper naming. | Weight: 15

### Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.